

實習生月報

PDO 資料路徑重建

CoE / SDO 交易重建

PDO Mapping / Assignment 關聯整理

SyncManager / FMMU 資料路徑分析

報告期間

2026.08

MONTHLY REPORT

本月工作大綱



08/xx - 08/xx

理解封包與 EtherCAT 100%

- 使用 Wireshark 觀察從上電抓的原始 EtherCAT 封包
- 整理 Broadcast、AP、FP 等指令對 Slave 的操作方式
- 整理 Master 對 Slave ESC 暫存器的讀寫流程
- 理解 CoE、SDO、PDO 與 Object Dictionary 的關係
- 建立本月主線：CoE / SDO → PDO Mapping / Assignment → SyncManager → FMMU

08/xx - 08/xx

建立 Slave 拓樸關係 80%

- 理解 Master 如何透過 Auto Increment / Configured Address 指定 Slave
- 理解 Configured Station Address 與 Configured Station Alias 的差異
- 設計 AP Datagram 配對、Slave Discovery 模型與分析流程

設計 Analyzer 驗證 Agent 20%

- 建立 Stage 3 Result Check
- 由 Capture evidence 驗證 Slave Count、Topology、Address 與 Identity

08/xx - 08/xx

Mailbox / CoE / SDO 70%

- 在 Datagram Record 中加入 CoE / SDO decoded fields
- 重建 SDO Initiate Download Request / Response Transaction
- 依 SDO Index / SubIndex 辨識 PDO Mapping / Assignment 操作
- 重建 RxPDO / TxPDO Mapping Configuration
- 重建 0x1C12 / 0x1C13 PDO Assignment Configuration

完善 Agent 驗證流程 30%

- 整理 TShark EtherCAT / CoE / SDO 查詢與 JSON 結構規格
- 將常用 Capture 查詢封裝成 Python tools，提供 LLM 多種驗證方式

08/xx - 08/xx

SyncManager / FMMU 分析 100%

- 在 Datagram Record 中加入 SyncManager / FMMU decoded fields
- 重建 SyncManager Configuration
- 重建 FMMU Configuration
- 驗證 PDO Assignment 與 SM2 / SM3 的設定關係
- 驗證 FMMU Physical Address 與 SyncManager RAM 區段的對應

01

EtherCAT Operation Flow

封包操作與資料路徑

從 Master 對 Slave 的封包操作，整理初始化、PDO 組態與週期資料交換主線

DISCOVERY — CONFIG — PROCESS DATA

Master → Slave Operations

01 STARTUP / DISCOVERY

APWR / APRD

Station Address

Slave Identity

02 PDO CONFIGURATION

SDO Download

PDO Mapping

SM Assignment

03 PROCESS DATA PATH

FPWR

SyncManager

FMMU

04 CYCLIC DATA

LWR / LRD / LRW

Process Data Exchange

Cyclic Data

Command	Data Path	Operation	Typical Use
LWR	Master → Slave Output Process Image	寫入輸出資料	常見於控制輸出更新
LRD	Slave Input Process Image → Master	讀回輸入資料	常見於狀態 / 感測資料回讀
LRW	Read + Write in one datagram	同時讀寫	雙向 cyclic exchange

「Master 寫入 Slave Output」是指主站把控制資料寫進 slave 的 output process image ；「Master 讀取 Slave Input」是指主站從 slave 的 input process image 取回狀態 / 感測資料。slave 進入 SAFEOP / OP 後，才進入這種週期性 process data 交換。



Startup Register Access

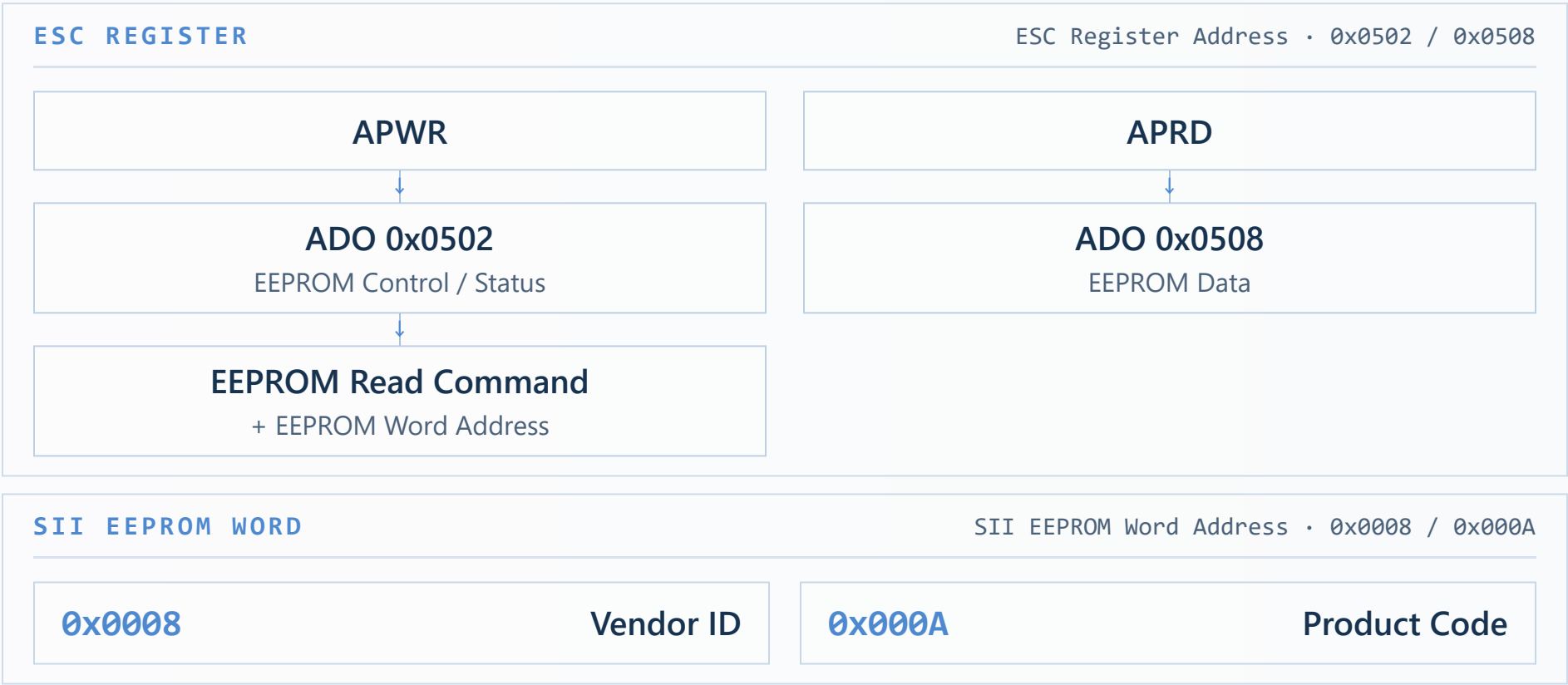
01 STATION ADDRESS
0x0010
Configured Station Address

02 AL STATE
0x0120 / 0x0130
AL Control / AL Status

03 SII EEPROM
0x0502 / 0x0508
Vendor ID / Product Code

SII EEPROM

ESC Register → SII EEPROM Word



Auto Increment → Slave Discovery

[STEP 01]

[STEP TITLE]

[STEP BODY]

[STEP 02]

[STEP TITLE]

[STEP BODY]

[STEP 03]

[STEP TITLE]

[STEP BODY]

[STEP 04]

[STEP TITLE]

[STEP BODY]

[STEP 05]

[STEP TITLE]

[STEP BODY]

[RESULT TITLE]

[RESULT ITEM]

[RESULT ITEM]

[RESULT ITEM]

[RESULT ITEM]

[RESULT ITEM]



分析問題與目標

01 PROBLEM

封包解析提供的是獨立的 **CoE / SDO Record**

單一 Datagram 無法直接表示一次完整的 SDO object access，
Request 與 Response 必須跨封包建立關聯。

- Request / Response 分散
- Index / SubIndex 資訊分散
- Result / Abort 狀態需要整合

02 GOAL

重建完整的 **SDO Transaction**

將 Mailbox / CoE / SDO 欄位重新組合，
建立可供後續 PDO Mapping 分析使用的 transaction-level model。

- Request ↔ Response
- Index / SubIndex
- Access Direction
- Result / Abort

INPUT

Mailbox / CoE Datagram

PROCESS

Transaction Reconstruction

OUTPUT

SDO Transaction Model



SDO Transaction Reconstruction Flow



EtherCAT Spec 重點整理

01 KEY CONCEPT	Core concept or mechanism Summarize the core mechanism, state transition, or protocol rule that defines the behavior described in the specification. Note what the device or analyzer should observe when this rule is applied.
02 IMPORTANT FIELDS	Registers / objects / parameters Record register addresses, object indices, data types, access rules, and parameter constraints. Highlight which fields are read-only, writable, or dependent on operating state.
03 DATA FLOW	How data moves through the system Trace how the request, datagram, mapped memory, and application data move through the system. Mark the boundaries where timing, ownership, or synchronization affects the result.
04 ENGINEERING NOTE	Implementation or analyzer relevance Connect the specification detail to source code, packet analysis, validation steps, or observable system behavior. Record the implementation risk or evidence that should be checked.



EtherCAT 圖解分析

FIGURE / 04



[Diagram Canvas]

Class Diagram / Architecture Diagram / Data Path Figure

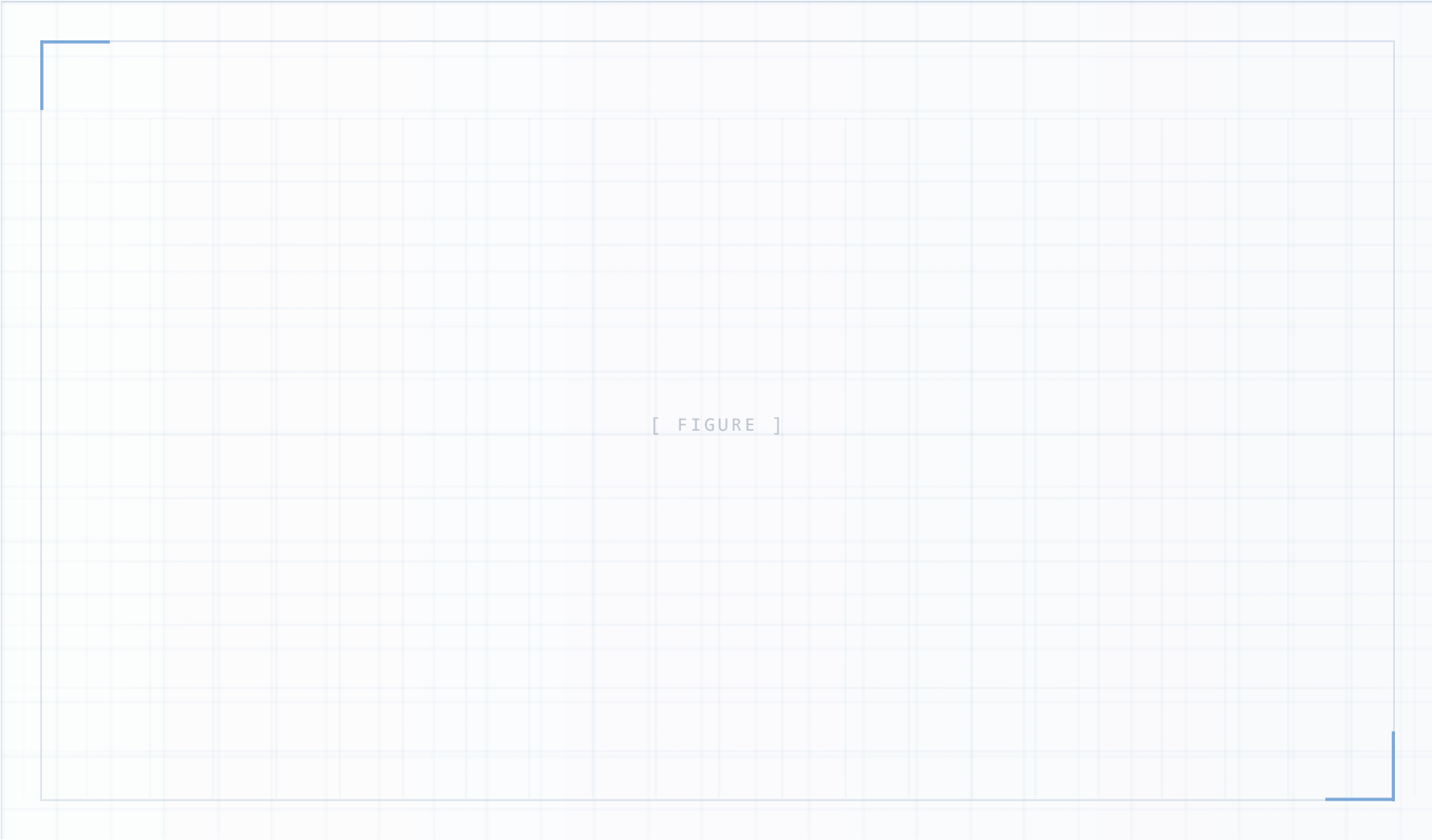
REPLACE WITH IMG / SVG / FIGURE / HTML DIAGRAM

FOCUS POINTS

- 01 Diagram Purpose**
Explain the main relationship between modules.
- 02 Key Components**
Highlight the major classes or architecture blocks.
- 03 Reading Order**
Describe how to read the diagram efficiently.
- 04 Implementation Relevance**
Connect the diagram to the current development work.



EtherCAT 資料路徑概覽



KNOWLEDGE POINTS

01

FMMU

Logical Address
→ ESC local

02

SyncManager

Consistent data
exchange control

03

Process RAM

Stores mapped
process data

04

DATA PATH

Logical Address → FMMU →
SM → RAM → Application